Kernel Moments and Sampling Error in the Joint Dickman Law

Source-Specific Reformulations and Local Refinements

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Abstract

We study selected analytic and probabilistic steps in the September 24, 2026 manuscript *The joint Dickman law for consecutive integers*. The fair-split channel, its two-split kernel, the latent row moments, and the adaptive cut comparison are kept in their original measures and normalizations. We prove a quadratic coarse-mesh bound for the positive smoothed channel kernel, a total-variation-to-operator estimate, exact same-site Laplace-feature and mixed row-moment identities, and a constructive positivity-preserving feature approximation. A finite sampling theorem records separately the skeleton, concentration, and exceptional-event costs and explains a sufficient abstract exponent region. These deductions use actual scoped private moment proofs and the public source, but do not independently certify its complete arithmetic argument, improve the joint Dickman theorem, or supply a number-theoretic convergence rate. Contribution assessment: C2/5, because the proved local refinements sharpen feature and sampling estimates without new arithmetic cancellation or a stronger global theorem.

1 Scope, provenance, and mathematical target

The source [1] asserts that for every fixed $a, b \in (0,1)$, through all real $X \to \infty$,

$$\frac{1}{X}\#\{2 \le n \le X : P^+(n) \le n^a,\ P^+(n+1) \le n^b\} \longrightarrow \rho(1/a)\rho(1/b). \tag{1}$$

Here $P^+(1) = 1$, $P^+(n)$ is the largest prime factor for $n > 1$, and the continuous Dickman function satisfies $\rho = 1$ on $[0,1]$ and $u\rho'(u) = -\rho(u-1)$ for $u > 1$. This is ordinary counting, not logarithmic density, and is an all-cutoff assertion for fixed parameters, not a statement uniform over growing parameter families. The second threshold is n^b, as in the source. Both coordinates in its ordering argument use the same denominator $\log n$.

The present report addresses the intermediate kernel and sampling mechanism. "Source" means an inherited assertion or a reconstruction of a source deduction. "Local" means a theorem proved here under its displayed hypotheses. "Conditional" means that applying such a theorem to the arithmetic problem retains the stated source inputs. No local theorem below replaces the prime-product laws, the weighted short-interval theorem, or the representation-multiplicity sieve. An applicability rating L3 describes relevance of the private methods; it is not the demonstrated-contribution rating C2.

The parent authenticated equality of the source subtree between the following pinned and captured October 8 commits:

```
adc7f1241b42e322a6451854ab7e4b4c146bf78a
fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb
```

This is snapshot provenance, not a live branch check or a proof audit. The author independently recorded pre-read hashes of the 13 assigned TeX files and three selected private originals; the final ledger checks their endpoint hashes and records literal read intervals and omissions. The public bibliography was separately displayed. No confidential original or report text was submitted to public search.

1.1 Whole-source architecture and scoped reading

All files in the following table were structurally mapped. File names refer to `build/sections/`, except the two front files. "Full" refers only to literal reading of that file, not verification of its external inputs. The exact intervals are retained in `M024-read-scope.json`.

File	Role in the source	Author reading
`main.tex`, `preamble.tex`	Assembly and notation; all eleven scientific sections are linked.	Structure only.
`introduction.tex`	Ordinary joint law, prior work, proof dependencies.	Full.
`labels.tex`	Bin counts, marginal factorial moments, centered labels, weighted short averages, bad-sequence profile.	Lines 201–337; other proofs imported.
`arithmetic.tex`	Moving roughness cutoff, local laws, reducing-weight sieves, coefficient moments, regularity losses.	Lines 1–145 and 278–357; remaining proof chain imported.
`amplification.tex`	Positive divisor amplifier, one- and two-split moments, preserved correlation, positive energy.	Structure only; results imported.
`graph.tex`	Change of variables, block boundary loss, site/root coupling, conditional row means.	Lines 145–480; earlier change-of-variable proof imported.
`moments.tex`	Tilted first representation, entropy and harmonic sieve, multiplicity, row squares.	Full; sieve inputs remain imported.
`channels.tex`	Two splits of one site, residue flattening, smoothed common-product kernel, coarse compactness.	Full.
`kernel.tex`	Regularity removal, arcs, histograms, residue factor, narrow-band inversion, finite features.	Full; arithmetic estimates imported.
`sampling.tex`	Integrated moments, column skeleton, conditional means, good-good extension, adaptive cut bound.	Full.
`conclusion.tex`	Feature polynomials, periodic lag approximation, block factorization, ordered parameter choices.	Full.
`distribution.tex`	Finite Fourier inversion, Dickman marginal, all real endpoints, moving thresholds, ordering.	Full; initial factorial moments imported.

This map deliberately leaves the amplifier and substantial arithmetic and label proofs outside the literal-read scope. In particular, a read of the application of a Selberg–Delange, character, sieve, or short-interval estimate is not an independent proof of that estimate. No entire-family, whole-corpus, or formal-kernel audit is asserted.

2 The source mechanism with its normalizations

Fix the source's regularity grid and tolerances. Its auxiliary scale is

$$T = \lfloor B^{0.32} \rfloor, \quad P_0 = B^{1000}, \quad \mathcal{P} = \{p : P_0 < p \le e^{4B}\}, \quad R = B/\log P_0, \quad M = \lceil C_6 T \rceil. \tag{2}$$

The probability space Ω_B consists of subsets $S \subset \mathcal{P}$, with independent inclusion probabilities $1/p$. Given S, a fair split selects each of its primes with probability $1/2$; write b for their product and $Y = \log b/B$. The unconditional product law is $\nu_{1/2}$. The untruncated source weights satisfy the exact identity

$$A_0(b)K_0(S \setminus b) = B2^{-|S|}, \qquad A_0(v) = (\log P_0)R^{1/2}\mu^2(v)2^{-\omega(v)}\mathbf{1}_{\{v \text{ rough}\}}. \tag{3}$$

Here $K_0(S) = R^{1/2}2^{-|S|}$, and μ in this formula is the Möbius function. Regularity restrictions stay as indicators inside expectations. One must not divide by the probability of an admissible split, thereby changing the weights.

Let $I = [1/2, 3]$ carry Lebesgue measure, not probability measure. Partition it into m_1 equal coarse cells I_l and a refining fine partition of mesh $\delta_B \asymp B^{-11/10}$. For $1 \le d \le B$, the residue space $(\mathbb{Z}/d\mathbb{Z})^\times$ carries uniform probability measure. The source channel is

$$U_d g(x, r) = \frac{\varphi(d)}{\delta_B}\mathbb{E}[g(S)\mathbf{1}_{\{Y \in I_x^{\text{fine}},\ b \equiv r \pmod d\}}], \quad U_d : L^2(\Omega_B) \longrightarrow L^2(dx\, d\mathrm{unif}(r)). \tag{4}$$

Writing P_{av} for residue averaging, $P_{\text{av}}U_d = U_1$. The coarse feature and its exact dual identity are

$$V_l(S) = |I_l|^{-1}\mathbb{P}(Y \in I_l \mid S), \qquad (P_{m_1}U_1 g)|_{I_l} = \mathbb{E}[g(S)V_l(S)]. \tag{5}$$

Thus $0 \le V_l \le |I_l|^{-1}$, and the identity allows every L^2 input, including inputs depending on B, position, and fixed external conditioning. It is not a claim that finitely many features reconstruct every element of $L^2(\Omega_B)$.

An allowed edge has $\eta T \le |k - i| \le T$ and distinct endpoints. The source latent kernel $\mathcal{L}_{ik}$ is the nonnegative sum over regular candidate triples $a - b = (k - i)c$ with endpoint subset products $b \subset S_i$, $a \subset S_k$, coefficient weights and complementary site weights, multiplied by k_B/B, where $k_B = \mathbb{E}K(S)$. After Fourier detection, residue averaging, and narrow-band inversion, its comparison kernel is

$$\mathcal{M}_{ik} = a_{ik}\sum_l h_l V_l(S_i)V_l(S_k), \tag{6}$$

$$a_{ik} = \frac{k_B}{T}\mathfrak{S}(|k - i|)w(s, |k - i|/T), \qquad h_l = \int_{I_l} m(y)\rho_0(y)\, dy \ge 0,$$

$$\mathfrak{S}(j) = \frac{j}{\varphi(j)}\prod_{p \nmid j}\left(1 - \frac{1}{(p-1)^2}\right), \quad j \ge 1. \tag{7}$$

All matrices remain zero on nonpairs. The singular series is zero for odd j and bounded above by $j/\varphi(j)$. The smooth weight has the source's nonsingular representation

$$w(s, \lambda) = \int_0^\infty \frac{\Psi_s(u + \lambda, u)}{u(u + \lambda)}\, du. \tag{8}$$

The support of Ψ_s keeps the denominators away from zero, so this extends smoothly to $\lambda = 0$. In the preceding inversion $v = B(x - y)$, $dx = dv/B$ cancels the prefactor B, and $1/j = 1/(T\lambda)$. There is no additional Fourier-inversion Jacobian.

The source's real cut norm is

$$\|H\|_\square = \frac{1}{M}\max_{\epsilon_i, \delta_k \in \{-1,1\}}\left|\sum_{i,k} H_{ik}\epsilon_i\delta_k\right|. \tag{9}$$

It is not normalized by M^2. Deleting indices retains normalization M. For $\mathcal{E} = \mathcal{L} - \mathcal{M}$, the source first proves an integral comparison over arbitrary bounded single-site tests,

$$\sup_{|g_i|, |h_k| \le 1}\frac{1}{M}\left|\sum_{i,k}\mathbb{E}[\mathcal{E}_{ik}g_i(S_i)h_k(S_k)]\right| \le Ce^{-cC_*} + C_\eta\epsilon_1 + o(1). \tag{10}$$

This is not yet an expected adaptive cut bound: the maximizing signs in (9) can depend on all sampled sites.

The arithmetic bottleneck is the source estimate

$$\mathbb{E}[\sum_k \mathcal{L}_{ik} \mid S_i] \le C_0 \quad \text{for every type,} \qquad \sum_k \mathbb{E}\mathcal{L}_{ik}^2 \ll B^{-0.21}. \tag{11}$$

The second bound is integrated over S_i, not pointwise conditional. Its proof tilts each unforced prime to probability $1/(2p-1)$ after selecting the first representation. The regularity indicators remain inside the tilted expectation. Prefix entropy, the harmonic $1/|j|$ progression gain, and the two-root reducing sieve bound the weighted alternative multiplicity by $O(B^{0.009})$. Combined with the five-factor weight exponent $\kappa_1 = 1 - 5(1 - \log 2)/2$, this yields the row-square estimate after summing the mean-bounded lag envelope. These arithmetic estimates are imported here even though their use and the complete multiplicity text were read.

3 Private proof routes and legitimate transfers

Three actual private originals, not just catalogue descriptions, were read in the following bounded intervals. Their complete local paths and pre/post SHA-256 values are in the confidential audit ledger. The short names below are local identifiers, not public citations.

1. `ch03-local-moments-rev2`, lines 1–226: the local Taylor integral remainder, output/source localization, raw versus central second moments, and polynomial exactness versus closure. Its two embedded chapter inputs were not followed or counted as read.

2. `TR-rev-1-old-tr250`, lines 121–181 and 455–477: physical spaces before coordinates, compatible finite moment banks, and the positive weighted polynomial Gram/unisolvence argument.

3. `TR-core-pat-16`, lines 184–374: a single joint law, exact covariance by common features, raw mixed moments, and the L^2 remainder bound with its cross-covariance terms.

The first proof applies Taylor expansion to the *observable* under an explicitly indexed kernel. It does not justify differentiating an arbitrary bounded endpoint test $g(S)$ or the cell indicator in (5). Here smoothness is used only for the source's cut density or for an explicitly smoothed feature. The second route guarantees positive finite Gram matrices under an injective polynomial evaluation map; it does not provide an inverse of the infinite-dimensional channel. The third route requires one common law for all feature products. It is precisely this requirement that distinguishes two independent splits of the *same* site from two independent sites below.

These are classical finite-dimensional and approximation principles present in the private originals. Their use is credited as inherited methodology, not as proof that the private corpus establishes a joint Dickman law. The actual local improvements are proved next.

4 Positive kernel compression and a sharper mesh bound

The source's smoothed common-product kernel is particularly well suited to the common-feature method. Fix its exclusive-product cutoff $0 < \theta < 1/20$ and let $f_{B,\theta}$ denote the smooth cut version of $f_{1/4,B}$, extended by zero to $\mathbb{R}$. It has support in a fixed compact interval and uniformly bounded derivatives for this fixed θ. If $Z = \log C/B$, where C has the independent common-product law $\nu_{1/4}$, put

$$f_Z(x) = f_{B,\theta}(x - Z), \qquad K_{B,\theta}(x,y) = \mathbb{E}[f_Z(x) f_Z(y)], \quad x, y \in I. \tag{12}$$

The law of Z may vary with B, have atoms, and have no density. No derivatives of that law occur in the following argument.

Theorem 4.1 (Local quadratic coarse-mesh bound)**.** *Let I be a finite interval, partitioned into cells J_l of lengths $h_l > 0$, and let P be cell averaging on $L^2(I, dx)$, $Q = 1 - P$. Suppose $f_Z \in L^2(I)$ is a real random function with $\mathbb{E}\|f_Z\|_2^2 < \infty$. The operator $\mathcal{T} = \mathbb{E}[f_Z \otimes f_Z]$ is positive and*

$$\|Q\mathcal{T}Q\| \le \mathbb{E}\|Qf_Z\|_2^2. \tag{13}$$

If $|f_Z(x) - f_Z(y)| \le D_l(Z)|x - y|$ for $x, y \in J_l$, then

$$\|Q\mathcal{T}Q\| \le \frac{1}{12}\sum_l h_l^3\, \mathbb{E}D_l(Z)^2. \tag{14}$$

In particular, equal cells of mesh $h = |I|/m_1$ and a uniform Lipschitz constant D_θ give $\|Q\mathcal{T}Q\| \le |I|D_\theta^2 h^2/12$.

Proof. For $u \in L^2(I)$, $\langle Q\mathcal{T}Qu, u\rangle = \mathbb{E}|\langle Qf_Z, u\rangle|^2 \le \mathbb{E}\|Qf_Z\|_2^2\|u\|_2^2$. For an integrable square-integrable function f on a cell J of length h,

$$\int_J |f - \overline{f}_J|^2 = \frac{1}{2h}\int_J\int_J |f(x) - f(y)|^2\, dx\, dy.$$

For a D-Lipschitz function this is at most $D^2h^3/12$, since the double integral of $(x-y)^2$ is $h^4/6$. Sum the cells and take expectations. These computations also prove the claims for random cellwise Lipschitz constants with the displayed finite second moments. □

For the source kernel (12), $D_l(Z)$ may be replaced by $D_\theta = \sup_B \|f'_{B,\theta}\|_\infty < \infty$. Its source comparison with $U_1U_1^*$ and its small-exclusive-product estimate therefore give the following conditional application:

$$\|(1 - P_{m_1})U_1\|^2 \le C(2\theta + 1/R)^{1/4} + \frac{|I|^3 D_\theta^2}{12m_1^2} + o_{\theta,m_1}(1). \tag{15}$$

Indeed the refining fine projection commutes with P_{m_1} and is a contraction, so it does not enlarge the compressed positive kernel. The source bounds the full difference $\mathcal{T} - P_{m_1}\mathcal{T}P_{m_1}$ by $O_\theta(m_1^{-1})$. The improvement here is for the *doubly compressed* residual $Q\mathcal{T}Q$, the quantity actually needed for $\|QU_1\|^2$. It does not assert an $O(m_1^{-2})$ bound for that full difference.

For fixed θ the mesh part of the channel error is now $O_\theta(m_1^{-1})$, rather than $O_\theta(m_1^{-1/2})$. Choose θ first to meet the small-product budget, then m_1, then B. No uniform bound on D_θ as $\theta \downarrow 0$ is claimed. There is consequently no automatic global rate in B. The nonuniform version (14) also permits smaller cells where a certified derivative envelope is larger; the source's refinement and boundary arguments would then have to use that fixed partition consistently.

Lemma 4.2 (Local probability-to-operator conversion)**.** *Let two joint probability laws on an output space, with total-variation distance at most ε, be recorded on disjoint cells of equal reference measure $\mu > 0$. Restriction to a subset of output cells is allowed without renormalizing. If their step kernels are the joint cell masses divided by μ^2, their operator difference on the reference L^2 space is at most $2\varepsilon/\mu$.*

Proof. Write Δ_{ab} for the joint cell-mass difference. With TV defined as half the total absolute signed mass, $\sum_{a,b}|\Delta_{ab}| \le 2\varepsilon$. Each absolute row or column integral of the step-kernel difference is at most $2\varepsilon/\mu$. The two-sided Schur bound proves the claim. This uses the total variation of the joint law, not merely the probability of a bad marginal event. □

The source's coupling of two splits to independent common and exclusive products has $\varepsilon = O(P_0^{-1})$ and $\mu = \delta_B/\varphi(d)$. Thus its coupling contribution can locally be bounded by $O(B^{21/10}/P_0)$ for $d \le B$, instead of its sufficient crude $O(B^5/P_0)$. At $d = 1$ it is $O(B^{11/10}/P_0)$. Both are already negligible for $P_0 = B^{1000}$. This refinement does not replace the two-sided estimates for small exclusive products, whose total probability alone is not the required operator bound.

5 Exact split features and common-law mixed moments

For $v \geq 0$ define the Laplace feature

$$G_v(S) = \mathbb{E}[e^{-vY} \mid S] = \prod_{p \in S} a_p(v), \qquad a_p(v) = \frac{1 + p^{-v/B}}{2}. \tag{16}$$

The empty product is one, and $0 < G_v \leq 1$. At the actual integer site $S_i(n)$ this is exactly the source weight $G_{B,v}(n+i)$, using distinct primes, regardless of their multiplicities.

Proposition 5.1 (Local raw and centered feature tables)**.** *For any finite real nonnegative list $v_0, \ldots, v_d$, put $m_v = \mathbb{E}G_v$ and $R_{vw} = \mathbb{E}[G_v G_w]$. Under the source site law,*

$$m_v = \prod_{p \in \mathcal{P}} \left(1 - \frac{1 - a_p(v)}{p}\right), \tag{17}$$

$$R_{vw} = \prod_{p \in \mathcal{P}} \left(1 - \frac{1 - a_p(v)a_p(w)}{p}\right). \tag{18}$$

The matrices R and $R - mm^{\mathsf{T}}$ are positive semidefinite. For a real coefficient matrix A, the approximating feature vector $\widetilde{V} = AG$ has exactly mean Am and covariance $A(R - mm^{\mathsf{T}})A^{\mathsf{T}}$, with no independence assumption between its components.

Proof. At a prime the site is absent with probability $1 - 1/p$, or present with probability $1/p$ and contributes $a_p(v)$ to G_v. Independence over primes proves both products. The two positive semidefinite assertions are respectively $\mathbb{E}(c^{\mathsf{T}}G)^2 \geq 0$ and $\mathbb{E}(c^{\mathsf{T}}(G - m))^2 \geq 0$. Linear contraction gives the remaining identities. □

It is essential that generally $R_{vw} \neq m_{v+w}$. The former uses two conditionally independent fair splits of the *same* S; the latter uses one split in both factors. For one prime $p = 2$, $B = 1$, and $v = w = 1$, $m_1 = 7/8$, $R_{11} = 25/32$, and $m_2 = 13/16$. Thus $\mathrm{Var}(G_1) = 1/64$, while $\mathrm{Var}(e^{-Y}) = 3/64$. The missing $1/32$ is conditional split variance, not new site randomness. More generally, if $Z_v = e^{-vY}$ under one split, then

$$\mathrm{Cov}(Z) = \mathrm{Cov}(G) + \mathbb{E}[\mathrm{Cov}(Z \mid S)]. \tag{19}$$

This follows by subtracting conditional means and expanding the outer product; the cross terms have zero conditional expectation. Hence the single-split moment table overestimates the covariance matrix in the positive-semidefinite order, but is not an exact replacement. The constant feature has zero centered variance and must not be rejected for lack of positive definiteness.

5.1 A constructive bounded polynomial lift

The source uses qualitative polynomial approximation of smoothed cell indicators. The following explicit version makes degree, endpoint mass, positivity, and coefficient cost separate.

Theorem 5.2 (Local Bernstein–Laplace approximation)**.** *Let $J \subset [1/2, 3]$ be a cell of length $h > 0$, set $q^0(y) = h^{-1}\mathbf{1}_J(y)$ on $y \geq 0$, and choose $0 < \delta < \min(h/2, 1/10)$. There is a continuous piecewise-linear q_δ, valued in $[0, h^{-1}]$, equal to q^0 outside the δ-neighborhoods of its two endpoints, supported in $[1/2 - \delta, 3 + \delta]$, with Lipschitz constant at most $1/(h\delta)$. Let $\beta_B(\delta)$ be the $\nu_{1/2}$ probability of those neighborhoods. For every integer $n \geq 1$ there is a polynomial $p_n(t) = \sum_{v=0}^{n} \alpha_v t^v$, independent of B, such*

that

$$0 \le p_n(t) \le h^{-1} \quad (0 \le t \le 1), \tag{20}$$

$$\sum_v |\alpha_v| \le h^{-1} 3^n, \tag{21}$$

$$\|V_J - \sum_{v=0}^{n} \alpha_v G_v\|_{L^2(\Omega_B)} \le \frac{\sqrt{\beta_B(\delta)}}{h} + \frac{e^{3.1}}{2h\delta\sqrt{n}}. \tag{22}$$

Here $V_J = \mathbb{E}[q^0(Y) \mid S]$ and q_δ, p_n use the fixed cell and fixed δ, not the site law.

Proof. Linear ramps across the endpoint neighborhoods give q_δ. The function $f(t) = q_\delta(-\log t)$ on $(0, 1]$, with $f(0) = 0$, is Lipschitz on $[0, 1]$ with constant $L = e^{3.1}/(h\delta)$: it vanishes for $t < e^{-(3+\delta)}$ and the chain rule on each remaining linear piece gives the bound. Take the Bernstein polynomial

$$p_n(t) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} t^k (1-t)^{n-k}.$$

Its nonnegative weights sum to one, proving (20). For $K \sim \mathrm{Bin}(n, t)$, $|p_n(t) - f(t)| \le L\mathbb{E}|K/n - t| \le L\sqrt{t(1-t)/n} \le L/(2\sqrt{n})$. Expanding the powers of $1 - t$ gives $\sum_v |\alpha_v| \le h^{-1} \sum_k \binom{n}{k} 2^{n-k} = h^{-1} 3^n$. The smoothing error in $L^2(\nu_{1/2})$ is at most $h^{-1}\sqrt{\beta_B(\delta)}$. Finally conditional expectation is an L^2 contraction, and $\mathbb{E}[p_n(e^{-Y}) \mid S] = \sum_v \alpha_v G_v(S)$ exactly. □

Under the source interval law, the marginal density is bounded on these endpoint neighborhoods, so $\beta_B(\delta) \le 4C\delta + O(B^{-80})$ for the fixed cell. This is an imported marginal estimate, not an assumption of an exact continuous density for the finite prime-product law. Choose δ first, n second, and then B to achieve any prescribed feature tolerance. The approximant stays in $[0, h^{-1}]$ even though its monomial coefficients can have signs. Its degree and ℓ^1 coefficient cost can be large. The source short-average argument permits fixed coefficients; it supplies no uniform rate for an arbitrarily growing degree or coefficient norm.

For a vector of approximants the common-law covariance identity in Proposition 5.1 applies. If $V = \widetilde{V} + r$, $\|r\|_{L^2} \le \varepsilon$, then

$$\|\mathbb{E}V - \mathbb{E}\widetilde{V}\| \le \varepsilon, \qquad \|\mathrm{Cov}(V) - \mathrm{Cov}(\widetilde{V})\| \le 2\sqrt{\mathrm{tr}\,\mathrm{Cov}(\widetilde{V})}\,\varepsilon + \varepsilon^2. \tag{23}$$

Centering is an orthogonal contraction and Cauchy–Schwarz controls the two cross-covariances, proving the formula. This is the actual private remainder principle specialized to the source features. The deterministic bound is not an independent additive noise variance.

6 Latent rows, mixed terms, and exceptional vertices

The vector $X(S) = (\sqrt{h_l} V_l(S))_l$ makes (6) an endpoint-feature factorization. The lag mask and weights still depend on both indices. Although the feature Gram kernel is positive semidefinite, the full masked matrix need not be. In fact, a nonzero real symmetric positive-semidefinite matrix cannot have every diagonal entry zero. Positivity of the feature bank therefore does not bypass the cut comparison.

Proposition 6.1 (Local exact mixed row identities)**.** *Let $X_1, \dots, X_M$ be independent copies of a real finite-dimensional random vector with $\mathbb{E}\|X\|^2 < \infty$. Write $\mu_X = \mathbb{E}X$, $R_X = \mathbb{E}XX^{\mathsf{T}}$, and $\Sigma_X = R_X - \mu_X \mu_X^{\mathsf{T}}$. For deterministic real weights a_{ik}, $a_{ii} = 0$, set $H_{ik} = a_{ik} X_i^{\mathsf{T}} X_k$ and $D_i = \sum_{k \ne i} H_{ik}$. Put*

$t_i = \sum_{k\neq i} a_{ik}$ *and* $u_i = \sum_{k\neq i} a_{ik}^2$. *Then*

$$\mathbb{E}[D_i \mid X_i] = t_i X_i^{\mathsf{T}} \mu_X, \tag{24}$$

$$\operatorname{Var}(D_i \mid X_i) = u_i X_i^{\mathsf{T}} \Sigma_X X_i, \tag{25}$$

$$\mathbb{E}D_i^2 = u_i \operatorname{tr}(R_X \Sigma_X) + t_i^2 \mu_X^{\mathsf{T}} R_X \mu_X, \tag{26}$$

$$\mathbb{E}H_{ik}^2 = a_{ik}^2 \operatorname{tr}(R_X^2), \quad k \neq i, \tag{27}$$

$$\mathbb{E}[H_{ik}H_{il}] = a_{ik}a_{il}\mu_X^{\mathsf{T}} R_X \mu_X, \quad i,k,l \text{ distinct}. \tag{28}$$

Proof. Conditioning on X_i makes the terms with distinct other endpoints independent. Their conditional means and variances give the first two equations. Averaging the conditional variance plus the squared conditional mean gives (26), using $\mathbb{E}[X^{\mathsf{T}}CX] = \operatorname{tr}(CR_X)$. Independence of X_i, X_k gives $\mathbb{E}(X_i^{\mathsf{T}} X_k)^2 = \operatorname{tr}(R_X^2)$. For two distinct neighbors, averaging them first gives $\mathbb{E}(X_i^{\mathsf{T}}\mu_X)^2 = \mu_X^{\mathsf{T}} R_X \mu_X$. All the integrals are finite by the stated second moment and independence; a fourth moment of a single X is not required. □

The cross terms in the last equation are typically nonzero. Summing entrywise second moments alone therefore does not compute the second moment of a row sum. The feature table of Proposition 5.1, with the coefficient and $\sqrt{h_l}$ factors, computes these quantities exactly for the polynomial comparison kernel. It does not compute the moments of the original representation sum $\mathcal{L}$ without an additional approximation estimate in the corresponding moment norm. A cut error alone does not give that estimate.

For the original source kernel the correct general principle is instead conditional independence across neighbors plus a conditional first-moment bound. Let $H_{ik} \geq 0$ be a symmetric endpoint envelope, with $H_{ii} = 0$ and independent site types. If for every i,

$$\mathbb{E}[D_i \mid S_i] \leq C_0, \quad D_i = \sum_k H_{ik}, \qquad \sum_k \mathbb{E}H_{ik}^2 \leq q, \tag{29}$$

then

$$\mathbb{E}\operatorname{Var}(D_i \mid S_i) \leq q, \qquad \mathbb{E}D_i^2 \leq C_0^2 + q, \qquad \mathbb{E}\left(\frac{1}{M}\sum_i D_i\right)^2 \leq C_0^2 + q. \tag{30}$$

The first assertion uses conditional independence, the second the variance identity, and the third Jensen across rows. No conditional uniform second-moment estimate is used.

Lemma 6.2 (Local expected vertex-deletion cost)**.** *Under* (29)*, let* $d > C_0$ *and delete every edge touching a vertex with* $D_i > d$*. For any matrix* E *with* $|E_{ik}| \leq H_{ik}$*, the expected cut-norm cost, with denominator still* M*, is at most*

$$\frac{2qd}{(d - C_0)^2}. \tag{31}$$

Proof. Write $Z_i = D_i - \mathbb{E}[D_i \mid S_i]$, so $\mathbb{E}Z_i^2 \leq q$. Then $\mathbb{P}(D_i > d) \leq q/(d - C_0)^2$ and Cauchy–Schwarz gives

$$\mathbb{E}[D_i \mathbf{1}_{\{D_i > d\}}] \leq \frac{C_0 q}{(d - C_0)^2} + \frac{q}{d - C_0} = \frac{qd}{(d - C_0)^2}.$$

Every deleted ordered edge is charged at most by twice the sum of the bad-vertex degrees. Divide by M and average over i. □

This expectation estimate does not require independence between the bad set and the matrix. It is useful for a separately declared truncation budget. It does not by itself justify bounded differences for a data-dependent trimmed matrix: changing a site can change which other vertices are trimmed. The sampling proof below retains the source's good-good extension rather than making that substitution.

7 A finite adaptive sampling theorem

The following theorem is a parameterized, source-relative version of the source sampling proof. The sufficient exponent criterion in Corollary 7.2 was also supplied by the parent during this stage and is credited as a parent local deduction. The finite statement and full argument below are the author's formulation. The classical skeleton, Lipschitz extension, and concentration mechanisms are not claims of historical novelty.

Theorem 7.1 (Local finite sampling ledger)**.** *Let $M \geq 2$ and let $S_1, \dots, S_M$ be independent random variables on finite spaces, not necessarily identically distributed. Let $E_{ik}(S_i, S_k)$ be real symmetric endpoint kernels, zero on the diagonal, dominated in absolute value by nonnegative symmetric H_{ik} satisfying* (29)*. Assume the integral bound*

$$\sup_{\substack{|g_i|\leq 1\\ |h_k|\leq 1}} \frac{1}{M} \left| \sum_{i,k} \mathbb{E}[E_{ik} g_i(S_i) h_k(S_k)] \right| \leq E_0, \tag{32}$$

where the functions are real measurable single-site tests and $E_0 \geq 0$. Choose $1 \leq k < M$, $d > C_0$, $\tau > 0$, $a > 0$, and set

$$p = \frac{Mq}{(d - C_0)^2} < 1, \qquad N = \binom{M}{k} 2^k, \tag{33}$$

$$b = E_0 + \sqrt{q} + Lp + a, \qquad L > \frac{E_0 + \sqrt{q} + a}{1 - p}, \tag{34}$$

$$\Delta = \min\left\{1,\ p + \frac{q}{\tau} + N \exp\left(-\frac{a^2 M}{8d^2}\right)\right\}. \tag{35}$$

Then

$$\mathbb{E}\|E\|_\square \leq b + 2\sqrt{\frac{M\tau}{k}} + \frac{2kd}{M} + \sqrt{C_0^2 + q}\sqrt{\Delta}. \tag{36}$$

The tests in the cut norm may depend on the complete realization.

Proof. Common good event and integrability. Conditional on S_i, different off-diagonal summands of $D_i = \sum_j H_{ij}$ are independent. Thus (30) holds, and Chebyshev gives $\mathbb{P}(D_i > d) \leq q/(d - C_0)^2$. A union bound gives

$$\mathbb{P}(\Omega_{\mathrm{deg}}^c) \leq p, \qquad \Omega_{\mathrm{deg}} = \{\max_i D_i \leq d\}.$$

Markov gives $\mathbb{P}(M^{-1} \sum_{i,j} E_{ij}^2 > \tau) \leq q/\tau$. Moreover $A = M^{-1} \sum_{i,j} |E_{ij}|$ satisfies $\mathbb{E}A^2 \leq C_0^2 + q$ by (30). These estimates also hold on every fixed induced subset with the original normalization M. Conditioning on outside site types does not alter the remaining independent laws.

Small-column skeleton. On the intersection of the two good events, fix an optimizing column-sign vector δ. Set $r_i = \sum_j E_{ij}\delta_j$. For a uniform k-subset J, the estimator $\widehat{r}_i = (M/k) \sum_{j\in J} E_{ij}\delta_j$ is unbiased and has variance at most $(M/k) \sum_j E_{ij}^2$. Since $|r| - r\,\mathrm{sgn}(\widehat{r}) \leq 2|r - \widehat{r}|$, some J has normalized sign loss at most $2\sqrt{M\tau/k}$. Use $\mathrm{sgn}(0) = 1$. Removing all pairs with an endpoint in J costs at most $2kd/M$. Therefore, with $R' = \{1, \dots, M\} \setminus J$,

$$\|E\|_\square \leq Z_{J,\sigma} + 2\sqrt{M\tau/k} + 2kd/M, \tag{37}$$

where σ is a sign assignment on J and

$$g_i(u) = \mathrm{sgn}\left(\sum_{j\in J} E_{ij}(u, S_j)\sigma_j\right), \qquad Z_{J,\sigma} = \frac{1}{M} \sum_{j\in R'} \left| \sum_{i\in R'} E_{ij} g_i(S_i) \right|.$$

This is an existence statement over a finite family of $N = \binom{M}{k}2^k$ assignments, not conditioning on the optimizer.

Conditional mean. Fix a deterministic J, σ and condition only on its site types. The g_i are now fixed single-site functions. For each $j \in R'$, let $a_j(v) = \sum_{i \in R', i \neq j} \mathbb{E}_{S_i}[E_{ij}(S_i, v)g_i(S_i)]$. Use $h_j(v) = \operatorname{sgn} a_j(v)$ in (32), setting both families to zero outside R'. It bounds the normalized sum of $\mathbb{E}|a_j(S_j)|$ by E_0. Conditional variance in the other endpoints and Cauchy–Schwarz bound the remaining mean absolute fluctuation by $\sqrt{q}$ per column. Symmetry supplies the integrated column-square bound. Consequently

$$\mathbb{E}[Z_{J,\sigma} \mid S_J] \leq E_0 + \sqrt{q} \tag{38}$$

uniformly in the conditioned values.

Good-good Lipschitz extension. Let $\mathcal{G}_{R'}$ be the configurations for which all induced H-degrees are at most d. Its conditional complement has probability at most $p < 1$, hence it is nonempty. If two good configurations differ on a set A' of coordinates, only ordered edges incident to A' can change, including changes of g_i. The reverse triangle inequality and the degree bounds at *both* configurations give

$$|Z(\mathbf{u}) - Z(\mathbf{v})| \leq \frac{4d}{M}|A'|.$$

This is a global good-good Hamming bound; no path through good configurations is presumed. Extend $\min(Z, L)$ from the good set by

$$\widetilde{Z}(\mathbf{u}) = \min\left\{L, \inf_{\mathbf{v} \in \mathcal{G}_{R'}} \left(\min(Z(\mathbf{v}), L) + \frac{4d}{M} d_H(\mathbf{u}, \mathbf{v})\right)\right\}.$$

The triangle inequality proves its Lipschitz property, and the good-good bound proves equality with $\min(Z, L)$ on the good set. It is measurable because the spaces are finite, and takes values in $[0, L]$. Equations (38) and the exceptional probability give $\mathbb{E}[\widetilde{Z} \mid S_J] \leq E_0 + \sqrt{q} + Lp$. Bounded differences [9] now yield

$$\mathbb{P}(\widetilde{Z} > b \mid S_J) \leq \exp(-a^2 M/(8d^2)).$$

By (34), $b < L$. On $\mathcal{G}_{R'}$, the events $Z > b$ and $\widetilde{Z} > b$ therefore agree.

One exceptional-event charge. Integrate the conditional exponential tail and union-bound it over the N deterministic assignments. The one full event $\Omega_{\deg}$ implies every induced good event at once. Only exponential extension tails are multiplied by N; the degree failure probability p is paid once. Outside a set of probability at most Δ, (37) bounds the cut norm by the first three terms of (36). On the exceptional set use $\|E\|_\square \leq A$ and Cauchy–Schwarz with $\mathbb{E}A^2 \leq C_0^2 + q$. This proves the result. □

Corollary 7.2 (Conditional abstract exponent region)**.** *Suppose the hypotheses of Theorem 7.1 hold uniformly for a family with $M \asymp B^m$, $q = O(B^{-\gamma})$, bounded C_0, E_0, and $m, \gamma > 0$. It suffices for asymptotic sampling transfer that $\gamma > m/2$. Specifically, choose positive d_*, s_*, e_* with*

$$m - \gamma - 2d_* < 0, \qquad 2d_* < s_* < e_* < \gamma, \qquad s_* < m. \tag{39}$$

Set $d \asymp B^{d_*}$, $\tau = B^{-e_*}$, and $k = \lfloor MB^{-s_*} \rfloor$. Then $\mathbb{E}\|E\|_\square \leq E_0 + o(1)$.

Proof. Choose $d_* > \max(0, (m - \gamma)/2)$ sufficiently close to that lower endpoint that $2d_* < \min(\gamma, m)$. Such a choice is possible when $\gamma > m/2$. Then choose $s_* \in (2d_*, \min(\gamma, m))$ and $e_* \in (s_*, \gamma)$. The degree failure is $O(B^{m-\gamma-2d_*})$ and the squared-mass failure is $O(B^{e_*-\gamma})$. The skeleton losses are $O(B^{(s_*-e_*)/2})$ and $O(B^{d_*-s_*})$. For each fixed $a > 0$ take a fixed sufficiently large clamp L. The logarithm of the number of assignments is $O(B^{m-s_*} \log B)$, while the exponential tail has exponent of order B^{m-2d_*}. Their strict gap makes the union tail vanish. Apply the theorem, then let $a \downarrow 0$. □

The source choices are $m = 0.32$, $\gamma = 0.21$, $d_* = 0.07$, $s_* = 0.18$, and $e_* = 0.20$. They satisfy (39). The region is sufficient for this proof, not necessary for all sampling methods. In particular it does *not* authorize increasing the source's M toward $B^{0.42}$. The row moment and conditional degree hypotheses must first be proved uniformly at the proposed new sample size, together with the other arithmetic and root-comparison bounds. Their possible dependence on M is not removed by changing notation.

8 Arithmetic assembly and the complete error ledger

The local results are useful only if their output norm matches the next step. The arithmetic energy has the source normalization

$$\mathcal{Q}_{B,x}^{\mathrm{main}} = \frac{1}{Tx} \sum_{n:\, n/(Tx)\in\mathcal{I}_s} \frac{1}{M} \sum_{i,k} \overline{F_x(n+i)} F_x(n+k) \mathcal{M}_{ik}(S_i(n), S_k(n); n/(Tx)). \tag{40}$$

Here $\mathcal{I}_s$ is a fixed compact subinterval of $(0,\infty)$, $|F_x| \le 2$, and F_x is the centered prime-bin label, not a multiplicative function in its own right. The source proves its weighted short-average input by a separate interpolation and short-interval argument. That input is not a consequence of a finite feature representation.

8.1 Feature error without label independence

Suppose V_l and $\widetilde{V}_l$ are bounded in absolute value by H_l and $\|V_l - \widetilde{V}_l\|_{L^2(\Omega_B)} \le \zeta_l$. For a deterministic symmetric lag array with $\sup_i \sum_k |a_{ik}| \le R_a$, the difference of the two feature energies, for any labels $|z_i| \le L_F$, is bounded pointwise by

$$\frac{2L_F^2 R_a}{M} \sum_i \sum_l h_l H_l |V_l(S_i) - \widetilde{V}_l(S_i)|. \tag{41}$$

Indeed expand $V_iV_k - \widetilde{V}_i\widetilde{V}_k$ by replacing one endpoint at a time, and sum the symmetric absolute row bounds. No independence of labels or endpoints is used. At fixed B, each error at $S_i(n)$ depends on finitely many residues. Ordinary progression counting, followed by $\|\cdot\|_1 \le \|\cdot\|_2$, makes the iterated upper-limit cost in (40) at most $2L_F^2 R_a |\mathcal{I}_s| \sum_l h_l H_l \zeta_l$. The factor $1/(Tx)$ and the fixed interval length are retained. The constructive approximants in Theorem 5.2 allow the same $H_l = |I_l|^{-1}$ as the exact features.

8.2 A uniform periodic-lag refinement

For fixed $P \ge 2$, let $\mathfrak{S}_P$ be (7) with both prime products restricted to $p \le P$, extended periodically to all integer lags with period $D = \prod_{p\le P} p$. In particular, its value at residue zero is the value of that finite product, not an omitted diagonal term. The original pair mask still removes the actual diagonal. The parent pointed out the following explicit refinement of the source's positive divisor-tail argument.

Proposition 8.1 (Local uniform lag-mean bound)**.** *Let $C_A = \prod_p (1 + 1/[p(p-1)]) < \infty$. For every integer $T \ge 1$ and real $P \ge 2$,*

$$\frac{1}{T} \sum_{1\le j\le T} |\mathfrak{S}(j) - \mathfrak{S}_P(j)| \le \frac{3C_A}{\lfloor P \rfloor}. \tag{42}$$

Proof. Put $N = \lfloor P \rfloor$, $a(j) = j/\varphi(j)$, and $a_P(j) = \prod_{p\le P, p\mid j} (1 - 1/p)^{-1}$. The omitted nondividing factors lie in $(0,1]$ and their loss is at most $\sum_{p>P} (p-1)^{-2} \le 2/N$. Thus

$$|\mathfrak{S}(j) - \mathfrak{S}_P(j)| \le a(j) - a_P(j) + (2/N) a(j).$$

The positive divisor expansion $a(j) = \sum_{d|j} \mu^2(d)/\varphi(d)$ bounds its mean by C_A. The mean of the omitted divisor terms is at most

$$\sum_{\substack{d\geq 1\\ \exists p>P:\, p|d}} \frac{\mu^2(d)}{d\varphi(d)} \leq C_A \sum_{p>P} \frac{1}{p(p-1)} \leq \frac{C_A}{N}.$$

The first inequality is a union bound in a positive convergent Euler product; the last follows by summing $1/[n(n-1)]$ over all integers $n > N$. Combine the bounds. □

This is a rate for the lag approximation alone, uniform in T. The primorial period D, feature degree, and constants in the short-average input remain fixed before B grows. It is not a rate for (1).

8.3 Errors and order of choices

The source-to-report interface can be written as the following ledger. Source constants are retained symbolically when their arithmetic proofs have not been independently quantified here.

Step	Error or required bound	Status
Amplification and diagonal	Positive lower energy c_5; diagonal $o(1)$ in the iterated limit.	Imported amplifier.
Lag and block endpoints	$C\eta + C/C_6$; denominator remains M.	Source graph reduction.
Actual sites and auxiliary roots	$o(1)$ expected cut error, including coupling failures and independent root fluctuations.	Imported arithmetic coupling estimates; scoped graph proof read.
Integral endpoint comparison	$Ce^{-cC_*} + C_\eta\epsilon_1 + o(1)$.	Source arc and residue construction.
Coarse channel projection	(15); fixed θ and mesh, plus source local-law errors.	Local mesh refinement, conditional application.
Adaptive sampling	(36); one degree exception, one squared-mass exception, union of exponential tails.	Local finite theorem using imported row hypotheses.
Polynomial features	(22), then (41); no label independence.	Local constructive bound; marginal endpoint mass imported.
Periodic lag and block freezing	$O(P^{-1})$ lag mean; then $O(\delta_{\mathrm{blk}}) + O(T^{-1})$ after fixed P.	Local lag bound; source block proof.
Weighted short averages	Every fixed polynomial monomial, fixed residue modulus, and shift interval with length tending to infinity.	Imported number-theoretic input.
Joint law and moving thresholds	Finite Fourier inversion; continuous marginal law and fixed-exponent squeeze.	Source conclusion, conditional on mixed decorrelation.

For clarity, the source short-average obligation is, for every fixed $v \geq 0$, D, residue r, and compact $\mathcal{I}_s$,

$$\lim_{B\to\infty} \limsup_{x\to\infty} \frac{1}{Tx} \sum_{n:\, n/(Tx)\in\mathcal{I}_s} \left| \frac{1}{L_B} \sum_{\substack{i\in J_B\\ i\equiv r\ (D)}} F_x(n+i) G_{B,v}(n+i) \right|^2 = 0, \tag{43}$$

for every consecutive shift interval J_B of length $L_B \to \infty$, with its starting point fixed at each B. At fixed B the modulus, all multipliers, shifts, and prime sets are finite when $x \to \infty$. There is no interchange of these limits with a simultaneous growth regime.

After fixing the features and P, divide the position block into a fixed number of consecutive intervals of lengths between $c_{\delta_{\mathrm{blk}},C_6}T$ and $\delta_{\mathrm{blk}}T$. Discard block pairs crossing the boundaries $|j| = \eta T$ or T. For each row they contain $O(\delta_{\mathrm{blk}}T + 1)$ positions, so bounded periodic weights give the stated normalized error. On the remaining pairs, (8) varies by $O(\delta_{\mathrm{blk}})$. The energy then factors into finitely many products of the averages in (43), with coefficients $L_aL_b/(MT)$ and fixed residue-pair weights. Cauchy–Schwarz in the origin variable makes each term vanish. Neither the representative scaled lags nor the coefficients need converge with B; boundedness suffices.

In the contradiction argument choose C_* first, then η, C_6, then ϵ_1, ϵ_2 and their coarse partition. Feature, periodic, and block tolerances are chosen subsequently, all before the inner x limit and the outer B limit. The source obtains

$$\limsup_{B\to\infty} \limsup_{x\to\infty} \frac{I_{2,x}}{BT} \le Ce^{-cC_*} + C\eta + C/C_6 + C_{\eta,C_6}\epsilon_1 + C'\epsilon_2.$$

Its lower bound contradicts this after the prescribed fixed choices. The sampling estimate is uniform for each fixed s in the compact range; no simultaneous event for uncountably many s is required. One integrates the uniform expectation bound. Bounded complex labels cost at most the source's factor 16 on converting the real cut bound, independently of the arithmetic labels.

Finally finite Fourier inversion of the fixed bin characters gives fixed-scale joint smoothness at rational exponents. Monotonicity and continuity extend to real exponents and all real cutoffs. Removing $n < \varepsilon X$, then squeezing n^a, n^b between fixed powers of X, gives (1). Its limiting marginal distribution is $D(t) = \rho(1/t)$ for $0 < t < 1$, extended by zero and one at the two ends. The second log coordinate can exceed one at finite n, but by at most $1/(n \log n)$. The continuous independent limiting law has zero diagonal mass and equal ordering probabilities $1/2$. These are source deductions, not new theorems established by the finite tests or feature tables in this report.

9 Public comparison, advantages, and limitations

Public-title searches were refreshed on October 8, 2026. The primary release page identifies the assigned September 24 preprint. Teräväinen's binary-correlation paper [2] gives a logarithmic framework and corresponding smooth-number applications. Tao–Teräväinen's version 2 of April 25, 2026 [3] describes quantitative correlations and almost-all-scale prime-factor-count applications. Yang's July 17, 2026 preprint [4] records a lower-density improvement for largest-prime ordering. Those are different quantifiers from the assigned source's all-cutoff joint law. The searches do not certify that no other relevant work exists or establish historical priority for the present elementary local estimates.

The short-interval inputs are associated with Matomäki–Radziwiłł [5] and the corrected version 3 of Matomäki–Radziwiłł–Tao [6]; the arXiv metadata explicitly records a corrected proof of Proposition A.3. The source sampling method is related to small-sample optimization [7], graph cut-metric sampling [8], and classical bounded differences [9]. Public web inspection here was abstract, bibliographic, and version scoped, not a fresh full-proof audit of those works. Their source-specific applications were read locally as recorded above.

Advantages. The compressed positive-kernel argument uses one derivative bound on the exclusive density and no smoothness of the common-product law. It improves the mesh power while retaining exactly the source coarse features. The common-site Euler products distinguish raw, centered, and conditional moments without Gaussian assumptions. The Bernstein lift gives a nonnegative bounded approximation and explicit endpoint and degree budgets. The finite sampling theorem exposes all

constants relevant to the transition from fixed single-site tests to adaptive signs and accommodates independent nonidentical finite site spaces.

Costs and limitations. Small-product removal still limits channel compactness, and D_θ may grow as θ decreases. Monomial coefficients can be poorly conditioned despite the bounded polynomial itself. Finite exact Gram positivity is not an arithmetic decorrelation estimate. Integrated row squares do not become uniform conditional squares. The latent representation multiplicity is not bounded merely by counting features. The sufficient sampling exponent region is conditional on row estimates at the actual chosen size. No new largest-prime ordering bound, fixed-parameter convergence rate, simultaneous growing-period theorem, or extension to other shifts is claimed.

Feasible next questions, not completed claims. A useful next step would quantify the cutoff dependence of D_θ and the prime-product local-law constants together, then optimize a fixed-tolerance feature budget. Another is to implement the common-site feature table in a stable Bernstein basis while explicitly propagating rounding and moment errors. Improving the sampling range arithmetically would require a new uniform multiplicity and row-moment analysis at that range. Any growing-degree or growing-period application must reprove (43) with the required uniformity. These are separate research obligations, not promised breakthroughs.

10 Verification and contribution assessment

The accompanying author test script performs bounded exact rational checks of split Euler products, same-site versus single-split moments, mixed row identities, cell variance constants, skeleton variance and sign-loss inequalities, positive polynomial coefficients, and exponent choices. Numerical operator checks supplement those identities. The script records its own cases and outcome in `M024-tests.json`; parent tests are recorded separately and are not counted as author-run tests. None of these finite experiments verifies an infinite arithmetic limit.

The evidence bundle includes the literal-read session, full structure map, explicit unread intervals, pre/post source hashes, public-literature scope, local proof audit, native compilation attempt, fallback build evidence, all-page visual record, contribution assessment, and a frozen-file manifest. Successful compilation, visual inspection, and hash consistency are presentation and integrity checks, not mathematical certification.

C2 is justified by the local mesh, moment, feature, deletion, sampling, and lag estimates actually proved. The source's global arithmetic theorem, classical approximation tools, and private foundational results are not credited as new contributions. The parent-supplied exponent region and lag-tail refinement are explicitly attributed. No historical novelty claim is made.

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